CS667 – Practical Data Science

Project #2: Machine Learning Regression Analysis

Predicting Total Sales Amount Using Ensemble Learning

1. Introduction

This report presents the results of Project #2 for CS667 Practical Data Science, which extends the Exploratory Data Analysis (EDA) from Project #1 into a predictive modeling exercise. The objective is to build regression models that predict Total Sales Amount (Revenue) from a retail sales dataset containing 1,000 transactions spanning January 2023 to January 2024.

Four regression models were trained and compared: Linear Regression (base model), Random Forest Regressor, XGBoost Regressor, and Gradient Boosting Regressor (ensemble models). The models were evaluated using R^2, MAE, MSE, and RMSE, with cross-validation for robustness.

2. Data Processing Steps

2.1 Dataset Overview

The dataset contains 1,000 retail transactions with 9 columns: Transaction ID, Date, Customer ID, Gender, Age, Product Category, Quantity, Price per Unit, and Total Amount. The data has zero missing values and zero duplicate rows. All numeric columns were already in proper format with no currency-string conversions needed.

2.2 Data Cleaning

- Dropped Transaction ID and Customer ID (unique identifiers, not predictive features)
- Verified Date column as datetime; confirmed all numeric columns are properly typed
- Validated that Total Amount = Quantity × Price per Unit for all 1,000 rows

2.3 Feature Engineering

Seven new features were derived from the Date column and Age:

- **Temporal features:** Month, DayOfWeek (0=Mon to 6=Sun), WeekOfYear, IsWeekend (binary), Quarter
- **Season:** Mapped from Month (Winter, Spring, Summer, Fall)
- **AgeBin:** Age grouped into 5 bins (18–25, 26–35, 36–45, 46–55, 56+)

2.4 Encoding

- **Label Encoding:** Gender (Female=0, Male=1)
- **One-Hot Encoding:** Product Category, Season, and AgeBin (with drop_first=True to avoid multicollinearity)

After preprocessing, the final dataset contained 1,000 rows and 18 features plus the target variable.

3. Chosen Models and Results

The data was split 80/20 (train/test, random_state=42). StandardScaler was applied for Linear Regression; tree-based models used unscaled data. Four models were trained:

Model	R^2 (Test)	MAE ($)	RMSE ($)	R^2 (Train)	CV R^2	Type
Random Forest	1.0000	0.00	0.00	1.0000	1.0000	Ensemble
Gradient Boosting	1.0000	0.00	0.00	1.0000	1.0000	Ensemble
XGBoost	0.9888	36.30	57.34	0.9965	0.9963	Ensemble
Linear Regression	0.8502	173.96	209.40	0.8523	0.8494	Base

Best Model: Random Forest Regressor (R^2 = 1.0000, MAE = $0.00, RMSE = $0.00). The 5-fold cross-validation confirmed perfect scores across all folds (CV R^2 = 1.0000, std = 0.0000), confirming the model generalizes well and is not overfitting.

The reason tree-based models achieve R^2 = 1.0 is that Total Amount is a deterministic product of Quantity × Price per Unit. Trees learn this multiplicative relationship perfectly through recursive partitioning, while Linear Regression (R^2 = 0.85) struggles because it fits additive, not multiplicative, relationships.

4. Business Insights

4.1 Top Predictive Features

Feature importance analysis across all ensemble models revealed a consistent ranking:

1. **Price per Unit (76.8%):** The dominant predictor. Higher-priced items (e.g., Electronics at $300–$500) naturally produce larger transactions.
2. **Quantity (23.2%):** The second driver. Customers purchasing 4 units generate 4× the revenue of single-item purchases.
3. **All other features (~0%):** Age, Gender, Product Category, temporal features, and season have negligible importance in the tree-based models.

4.2 Actionable Recommendations

4. **Pricing Strategy:** Price per Unit is the #1 revenue lever. Optimizing pricing tiers, running strategic promotions, and testing premium pricing across categories will have the highest impact.
5. **Volume Campaigns:** Quantity is the second key driver. Bundle deals, loyalty rewards, and "buy more, save more" promotions can directly boost Total Amount.
6. **Product Mix:** Electronics transactions generate higher revenue per transaction. Shifting marketing toward high-margin categories or cross-selling can increase average transaction value.
7. **Demographics Are Secondary:** Age and Gender show minimal predictive power. Revenue optimization should focus on what customers buy, not who they are.
8. **Seasonality is Weak:** Temporal features have near-zero importance, suggesting steady year-round demand. Product-focused strategies outweigh seasonal campaigns for this dataset.

5. Conclusion

This project successfully transitioned from exploratory data analysis to predictive modeling. The Random Forest Regressor was identified as the best-performing model, achieving perfect prediction accuracy on the retail sales dataset. The ensemble models consistently outperformed the Linear Regression baseline, demonstrating the power of tree-based methods for capturing non-linear feature interactions.

The feature importance analysis revealed that revenue is overwhelmingly driven by Price per Unit and Quantity—a finding that aligns with the deterministic relationship in the data (Total Amount = Quantity × Price). For practical deployment, Random Forest is recommended due to its perfect accuracy, robustness through bagging, minimal tuning requirements, and interpretable feature importance output.